

Abdullah Akgül

Curriculum Vitæ

I build probabilistic algorithms that make reinforcement learning agents learn faster and adapt more reliably. Three of my first-authored algorithms (MOMBO on offline RL, EPPO on non-stationary control, and DAIF on online control) each rank first in sample efficiency on their respective benchmarks, with publications at NeurIPS, ICML, ICLR, and TMLR. Beyond research, I have deployed a deep learning fraud detection system in production at Vakifbank and shipped open-source ML packages (ObjectRL, MOMBO, EPPO) with CI, unit tests, and full API documentation, adopted by the research community. I am seeking research scientist or machine learning engineer roles focused on sequential decision-making, robotics, or autonomous systems.

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 Abdullah Akgül

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EDUCATION

- **Ph.D. in Computer Science** University of Southern Denmark, Denmark Feb 2023 - May 2026
– Thesis: *Probabilistic Reinforcement Learning for Sample-Efficient Control* 
- **M.Sc. in Computer Engineering** Istanbul Technical University, Turkey (GPA: 4.0/4.0) Sep 2020 - Nov 2022
– Thesis: *Memory-based Approaches to Problems in Probabilistic Modeling* 
- **B.Sc. in Computer Engineering** Istanbul Technical University, Turkey (GPA: 3.7/4.0) Sep 2016 - Aug 2020

TECHNICAL SKILLS

Programming Languages: Python, C, C++, JavaScript, HTML/CSS

ML Frameworks & Libraries: PyTorch, JAX, TensorFlow, Keras, Scikit-learn, NumPy, SciPy, Pandas, OpenCV

RL Environments & Benchmarks: MuJoCo/Gymnasium, DeepMind Control Suite (DMC), D4RL, NeoRL, EvoGym, MaMuJoCo, JaxMARL, PettingZoo, PyBullet

Tools & Platforms: Git, Linux, Docker, ROS, MATLAB, LaTeX, Weights & Biases, Flask, Node.js

Languages: English (Fluent), Turkish (Native), Danish (Basic)

Areas of Expertise: Reinforcement Learning (Model-Based, Offline, Multi-Agent), Probabilistic Modeling, Bayesian Inference, Uncertainty Quantification, Deep Learning, Computer Vision, Contextual Bandits, Large Language Models

EXPERIENCE

- **Postdoctoral Researcher** Feb 2026 - Aug 2026
Department of Mathematics and Computer Science, University of Southern Denmark Denmark
– Extending probabilistic RL to multi-agent settings, focusing on non-stationary dynamics in cooperative and competitive environments; co-authored a paper accepted to UAI 2026 on non-stationary linear contextual bandits.
– Contributing to the ADIN Lab open-source ecosystem, including ObjectRL, a pip-installable deep RL package with CI, unit testing, and full API documentation.
- **Ph.D. Researcher** Feb 2023 - Jan 2026
Department of Mathematics and Computer Science, University of Southern Denmark Denmark
– Published at NeurIPS, ICML, ICLR, TMLR, UAI, and L4DC on probabilistic methods for sample-efficient reinforcement learning (ADIN Lab, supervised by Prof. Melih Kandemir); all three first-authored algorithms rank first on their respective benchmarks.
– Built and open-sourced pip-installable Python packages (MOMBO, EPPO, ObjectRL) with unit testing, continuous integration, and full API documentation, adopted by the research community.
– Delivered exercise sessions on core machine learning concepts for undergraduate and graduate courses (~40 students), providing individualized feedback on assignments.
– Mentored 2 MSc students through the full thesis lifecycle, supporting research design and implementation.
- **Research and Teaching Assistant** Mar 2021 - Jan 2023
Artificial Intelligence and Data Engineering, Istanbul Technical University Turkey
– Conducted research in computer vision and uncertainty quantification at the Vision Lab (supervised by Prof. Gözde Ünal); co-authored a paper at ICLR 2022 and contributed to NeurIPS Workshop 2022 and ICLR Workshop 2024.
– Teaching Assistant for Python Programming, Probability & Statistics, and Computer Architecture (100+ students per course).
– Mentored 1 MSc student through the full thesis lifecycle, resulting in a TMLR publication; supervised 3+ BSc students on capstone projects, producing 2 workshop papers at NeurIPS and ICLR.
- **Machine Learning Engineer** Aug 2020 - Sep 2020
R&D and Innovation, Vakifbank Turkey
– Built an end-to-end fraud detection pipeline in TensorFlow 2/Keras: queried a large internal database, automatically extracted and cleaned signature regions from scanned cheque pages, and constructed a labelled training dataset from scratch.
– Trained and deployed a Siamese CNN achieving 95% accuracy on internal data and 88% on the public CEDAR benchmark, generalizing to previously unseen signers without retraining; structured the codebase for handoff to the R&D team.
- **Research Intern** Aug 2019 - Sep 2019
Artificial Intelligence and Robotics Laboratory (AIRLab), Istanbul Technical University Turkey
– Implemented deep reinforcement learning algorithms (DDPG, SAC) for robotic manipulation tasks.
– Developed PyBullet simulation environments with MoveIt integration for inverse kinematics.

•Distributional Active Inference

2026

- Akgül, A., Baykal, G., Haußmann, M., ..., Kandemir, M., *International Conference on Machine Learning (ICML)*.   
- Delivered the first model-free integration of Active Inference into distributional RL by parameterizing quantile regression as Bayesian NIG inference, enabling principled uncertainty-driven exploration with no world model; achieved best average rank across all 19 continuous control tasks on three benchmark suites (EvoGym, DMC, DMC Vision), with up to +62% sample efficiency over the next-best baseline.

•Deterministic Uncertainty Propagation for Improved Model-Based Offline Reinforcement Learning

2024

- Akgül, A., Haußmann, M., Kandemir, M., *Advances in Neural Information Processing Systems (NeurIPS)*.   
- Diagnosed high-variance Monte Carlo Bellman targets as the primary instability in model-based offline RL and replaced sampling with deterministic progressive moment matching, yielding provably tighter suboptimality bounds and best convergence rate (AULC rank 1.2) across all 12 D4RL settings; advantage is most pronounced on medium-quality datasets (hopper-medium AULC 95.9 vs. 82.2 for MOBILE and 37.0 for MOPO).

•Overcoming Non-stationary Dynamics with Evidential Proximal Policy Optimization

2025

- Akgül, A., Baykal, G., Haußmann, M., Kandemir, M., *Transactions on Machine Learning Research (TMLR)*.   
- Demonstrated that an evidential NIG critic simultaneously solves both compounding failures of PPO under non-stationary dynamics (loss of plasticity and undirected exploration) through adaptive gradient scaling and UCB-based advantage estimation; achieved best AULC rank across 10+ non-stationary MuJoCo scenarios (rank 1.5 slippery, 1.7 paralysis) while maintaining higher effective rank and fewer dormant neurons over 15 seeds.

•Evidential Turing Processes

2022

- Kandemir, M., Akgül, A., ..., Unal, G., *International Conference on Learning Representations (ICLR)*.   
- Proved the Complete Bayesian Model (CBM) is the minimal structure for total calibration by combining global model uncertainty with local evidential uncertainty; ETP achieves the only simultaneous top performance on NLL (best on all 5 datasets), ECE (2.6–2.7% on image benchmarks), and OOD detection across both text and image benchmarks; designed and conducted all experiments as 2nd author.

•Bridging the Performance-Gap Between Target-Free and Target-Based Reinforcement Learning

2026

- Vincent, T., ..., Akgül, A., ..., D'Eramo, C., *International Conference on Learning Representations (ICLR)*.   
- Resolved the target-free vs. target-based RL tradeoff by sharing all backbone parameters and duplicating only the final linear head, reducing memory by up to 49% while matching or exceeding target-based performance across five settings: Atari (+6%), offline Atari (gap 26% → 6%), DMC Hard (full recovery), streaming RL (+10%), and language RL on Wordle (+5%); led the language RL experiments, applying the method to a GPT-2 backbone and demonstrating that the approach transfers to large language models.

•Continual Learning of Multi-modal Dynamics with External Memory

2024

- Akgül, A., Unal, G., Kandemir, M., *Learning for Dynamics and Control Conference (L4DC)*.   
- Introduced the first continual learning method for multi-modal dynamical systems: neural episodic memory with a Dirichlet Process prior replaces parameter transfer, enabling automatic mode discovery and catastrophic-forgetting-free adaptation without mode labels; outperforms the VCL-BSSM baseline on 4/5 datasets (NMSE) and 5/5 datasets (NLL), with a 26% NMSE reduction on Character Trajectories.

•Weighted Sequential Bayesian Inference for Non-Stationary Linear Contextual Bandits

2026

- Werge, N., Wu, Y.S., Akgül, A., Kandemir, M., *Conference on Uncertainty in Artificial Intelligence (UAI)*. 
- Derived a novel concentration inequality for Weighted Sequential Bayesian posteriors in non-stationary linear contextual bandits, yielding three algorithms (WSB-LinUCB, WSB-RandLinUCB, WSB-LinTS) with improved frequentist regret guarantees over WRLS-based baselines at equivalent computational cost.

•Meta Continual Learning on Graphs with Experience Replay

2023

- Unal, A., Akgül, A., Kandemir, M., Unal, G., *Transactions on Machine Learning Research (TMLR)*.  
- Designed a meta-continual learning framework for node classification on graphs combining MAML-style meta-learning with experience replay; improved average accuracy by 10–80% over baselines across three benchmarks (CiteSeer: 51.48%, Arxiv: 31.76%, Reddit: 67.66%) while reducing catastrophic forgetting by up to 78 percentage points.

•ObjectRL: An Object-Oriented Reinforcement Learning Codebase

2025

- Baykal, G., Akgül, A., Haußmann, M., Tasdighi, B., Werge, N., Wu, Y.S., Kandemir, M., *arXiv preprint*.   
- Designed ObjectRL, the first fully object-oriented deep RL codebase (9 algorithms, 5 MuJoCo environments) where encapsulation, inheritance, and polymorphism map directly to RL components; extending SAC to a new algorithm requires overriding ~5 lines of code, released as a pip-installable Python package with CI, unit tests, and full API documentation.

•The rest is available in Google Scholar: <https://scholar.google.com/citations?user=FZeaKPoAAAAJ>.

ACADEMIC SERVICE

- Reviewer:** Advances in Neural Information Processing Systems (NeurIPS) 2025 - Present
- Reviewer:** European Workshop on Reinforcement Learning (EWRL) 2025 - Present
- Reviewer:** IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025 - Present
- Reviewer:** IEEE Transactions on Neural Networks and Learning Systems (TNNLS) 2022 - 2023